

KIROVSKII, Russian Federation

WMO: 318780

Lat: 45.100N

Lon: 133.500E

Elev: 321

StdP: 14.53

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.2	-19.1	-28.3	1.1	-22.5	-24.1	1.5	-18.4	14.2	8.4	12.3	8.2	1.0	180	0.479

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	86.1	72.7	83.5	71.0	81.0	69.8	75.3	82.6	73.6	80.4	71.9	78.3	4.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.1	124.3	79.7	71.4	117.1	77.8	69.8	110.8	76.2	39.2	82.6	37.6	80.6	36.0	78.3	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.9	11.8	10.0	-27.4	90.8	5.6	2.4	-31.4	92.6	-34.7	93.9	-37.8	95.3	-41.8	97.0
			-27.6	77.8	5.5	1.7	-31.6	79.0	-34.8	80.0	-37.9	81.0	-41.9	82.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.3	-1.3	7.2	24.0	43.4	56.8	65.7	72.0	70.4	59.5	44.3	23.4	4.4		
	DBStd	26.95	8.86	9.83	10.65	8.31	6.79	5.61	4.42	4.91	6.41	8.58	12.17	9.95		
	HDD50	6283	1591	1199	806	232	16	0	0	0	7	217	800	1415		
	HDD65	9890	2056	1619	1270	647	268	58	3	12	186	642	1249	1880		
	CDD50	2378	0	0	0	36	226	472	681	632	290	40	1	0		
	CDD65	511	0	0	0	1	13	80	219	178	20	0	0	0		
	CDH74	3347	0	0	0	18	176	598	1407	1023	121	4	0	0		
	CDH80	752	0	0	0	2	34	131	366	207	12	0	0	0		
	WSAvg	4.1	3.0	3.6	4.8	5.3	4.9	3.9	3.6	3.5	3.7	4.6	4.5	3.5		
	PrecAvg	26.2	0.6	0.6	1.1	1.5	3.1	3.1	4.6	3.9	3.1	2.4	1.3	0.9		
(15) Precipitation	PrecMax	34.8	1.4	1.4	3.0	4.3	7.0	6.3	10.7	6.7	7.2	5.0	2.7	2.3		
	PrecMin	16.0	0.0	0.0	0.0	0.1	0.6	1.1	1.5	1.6	0.6	1.1	0.2	0.1		
	PrecStd	4.7	0.3	0.4	0.7	1.0	1.3	1.2	2.0	1.5	1.8	1.0	0.6	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.0	40.4	55.1	76.4	84.3	87.8	90.5	88.0	81.6	72.3	57.2	39.0		
		MCWB	27.4	33.3	42.3	55.2	62.7	70.8	75.1	75.4	68.1	57.2	46.3	36.0		
	2%	DB	23.9	35.4	48.5	69.2	79.0	83.7	86.8	84.8	77.4	67.1	50.3	31.9		
		MCWB	21.1	30.2	38.5	51.4	60.1	68.5	73.8	73.3	65.3	54.2	42.7	29.0		
	5%	DB	18.0	29.9	43.7	63.5	74.5	80.5	84.0	82.2	74.2	62.1	45.5	25.8		
		MCWB	15.6	26.1	35.9	48.5	58.0	67.6	72.7	71.9	63.6	51.4	39.4	23.3		
	10%	DB	13.3	24.3	39.4	58.3	70.2	77.2	81.3	79.6	71.1	58.2	40.4	20.4		
		MCWB	11.4	20.9	33.4	45.7	55.9	65.7	71.4	70.5	61.8	49.6	35.5	17.8		
	0.4%	WB	27.6	34.8	43.3	56.2	64.4	72.9	78.2	77.4	71.1	59.5	47.6	35.9		
		MCDB	30.6	39.0	52.5	74.5	78.9	83.3	86.4	85.0	77.2	68.4	55.5	38.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	21.2	30.4	39.3	52.2	61.7	70.7	75.9	75.3	67.9	56.1	43.6	29.2		
		MCDB	23.8	35.0	46.8	66.5	75.8	80.6	83.5	82.0	74.0	63.9	49.5	31.8		
	5%	WB	15.6	25.9	36.3	49.5	59.5	68.8	74.1	73.7	65.4	53.5	39.6	23.4		
		MCDB	17.7	30.0	42.7	62.0	71.9	77.8	81.5	79.7	71.9	60.8	45.1	25.9		
	10%	WB	11.4	20.9	33.7	46.5	57.4	66.9	72.4	72.1	63.3	50.6	35.8	17.9		
		MCDB	13.3	24.3	39.0	57.0	67.3	75.4	79.2	77.5	69.1	57.1	40.3	20.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	22.1	23.2	19.0	19.4	19.5	17.8	15.2	15.4	19.7	20.1	17.9	19.5		
		MCDBR	26.1	25.4	22.2	28.4	27.6	23.2	19.8	19.1	23.1	26.7	22.2	22.1		
		MCWBR	23.2	20.9	14.2	14.9	12.7	11.0	8.8	9.0	12.0	14.9	15.5	19.7		
		MCDBR	25.3	23.8	20.2	26.4	23.9	20.7	17.3	15.8	18.8	22.6	21.6	21.9		
	5% WB	MCWBR	22.8	19.8	13.5	14.5	11.8	10.6	8.5	8.2	10.6	14.1	15.9	19.7		
(40) Clear-Sky Solar Irradiance	taub		0.319	0.375	0.456	0.565	0.577	0.554	0.512	0.474	0.432	0.453	0.394	0.340		
	taud		2.039	1.879	1.789	1.695	1.735	1.873	2.042	2.145	2.170	1.962	2.008	2.006		
	Ebn at Noon		237	242	238	224	227	234	243	247	247	214	204	211		
	Edh at Noon		33	47	60	72	71	62	52	45	41	43	34	31		
(44) All-Sky Solar Radiation	RadAvg		669	1009	1351	1476	1652	1779	1600	1479	1295	911	625	526		
	RadStd		36	46	87	144	127	195	155	117	108	75	49	29		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page